

Dedebit (DECSI) — On-Premises Installation

GitHub + one-command install · Seqela Technologies

Dedebit (DECSI) — On-Premises Installation Guide

Audience: Dedebit Credit and Savings Institution (DECSI) technical / infrastructure team

Product: AI-powered Credit Intelligence (Loan Hub + Digital Apply + collateral + market)

Vendor: Seqela Technologies

Deploy model: On-premises (your data centre / private cloud)

Licensing: Signed **license key** with an **expiry date** (renewed by Seqela)

GitHub: https://github.com/habenkiros/decsi_loan

This pack is what IT needs to install, license, smoke-test, and hand over to operations.

Related end-user manuals (admin / staff / customers) live under `docs/user_manual/` and inside the app at `/hub/help/`.

Quick start (simplest path)

On a Linux server with Docker 24+, Compose v2, and Git:

```
git clone https://github.com/habenkiros/decsi_loan.git
cd decsi_loan
./scripts/install_decsi.sh
```

That one script creates `.env`, applies the evaluation license, starts Docker, migrates the database, and can create a superuser.

Short handout: `[INSTALL_DECSI.md](../../INSTALL_DECSI.md)` at the repo root.

Continue below only if you need manual steps, HTTPS tablets, backups, or hardening.

1. What you receive from Seqela

Item	Purpose
Application release (Git tag / zip / Docker images)	The software
This installation guide	Step-by-step bring-up
`LICENSE_KEY` (string starting with <code>`SEQLA1.`</code>)	Activates the product until the embedded expiry date
Optional: user manuals PDF	Training / ops

Current evaluation key (this pack): 7 days (expires **2026-08-26**).

After commercial agreement, Seqela will issue the **Year-1** license key for production.

You do **not** receive Seqela's private signing key. Only Seqela can issue or extend license keys.

2. Minimum server requirements

Resource	Minimum (pilot)	Recommended
CPU	2 vCPU	4 vCPU
RAM	4 GB	8 GB
Disk	40 GB SSD free	100 GB+ (document media grows)
OS	Ubuntu 22.04 LTS or RHEL 8/9 (x86_64)	Same

Software	Docker Engine **24+** , Docker Compose **v2** , Git	Same
Network	LAN to staff PCs; static IP or DNS name	NTP synchronized (required for MFA)

Ports

Port	Use
8000/tcp	Application HTTP (or map via reverse proxy)
5432/tcp	PostgreSQL — **localhost / Docker network only**
8443/tcp	Optional HTTPS overlay for field tablets

3. License key model (important)

3.1 How it works

1. Seqela issues a signed key for **Dedebit / DECSI** with:
 - Organization name and code
 - **Issue date** and **expiry date**
 - Feature flags (hub, Digital Apply, market, collateral, ...)
2. Dedebit IT places the key in `.env` as `LICENSE_KEY=...` (or in `deploy/license/license.key`).
3. The app verifies the signature **offline** (no phone-home) using Seqela's public key baked into the product.
4. While the key is valid, the product runs normally.
5. Hub shows a **warning banner** when ≤ 30 days remain.
6. After expiry, a short **grace period** (default **7 days**) still allows login so IT can install a renewal key.
7. After grace, the product **locks** until a new key is installed (status page `/license/` stays reachable).

3.2 Install the license

```
# Option A – .env
echo 'LICENSE_KEY=SEQLA1.xxxxx.yyyyy' >> /opt/decsi_loan/.env

# Option B – file (one line, no quotes)
mkdir -p /opt/decsi_loan/deploy/license
nano /opt/decsi_loan/deploy/license/license.key
```

Recommended production settings:

```
LICENSE_ENFORCE=True
LICENSE_GRACE_DAYS=7
```

Check status:

```
docker compose exec web python manage.py check_license
# or open https://<host>/license/
```

3.3 Renew before expiry

1. Contact Seqela with your `org_code` (e.g. `DECSI`) and desired new expiry date.
2. Receive a new `SEQLA1....` string.
3. Replace `LICENSE_KEY` (or `license.key`).
4. `docker compose restart web`
5. Confirm `/license/` shows **Valid** and the new expiry.

Do not edit the middle of the key — any change invalidates the signature.

4. Installation steps (Docker Compose — required path)

Step 1 — Prepare host

```
sudo apt update
sudo apt install -y git ca-certificates curl
# Install Docker Engine 24+ and Compose v2 per Docker docs for
docker --version
docker compose version
timedatectl status    # NTP should be active
```

Step 2 — Obtain the release

```
sudo mkdir -p /opt
cd /opt
# Use the URL / media Seqela provides, for example:
# git clone <seqela-release-url> decsi_loan
# or: unzip Dedebit_Loan_Hub_Release_YYYYMMDD.zip -d decsi_loan
cd decsi_loan
```

Step 3 — Environment file

```
cp .env.example .env
nano .env
```

Minimum required values:

```
SECRET_KEY=<long-random-string>
DEBUG=False
ALLOWED_HOSTS=localhost,127.0.0.1,<server-ip-or-hostname>
SITE_URL=http://<server-ip-or-hostname>:8000

DB_NAME=decsiloandb
DB_USER=decsiloandbuser
DB_PASSWORD=<strong-password>
DB_HOST=db
DB_PORT=5432
```

```
LICENSE_KEY=<paste-SEQLA1-key-from-Seqela>
LICENSE_ENFORCE=True
LICENSE_GRACE_DAYS=7
```

```
LOGIN_MAX_FAILED_ATTEMPTS=5
LOGIN_LOCKOUT_MINUTES=15
MFA_REQUIRED=True
MFA_TOTP_ISSUER=DECSI Loan Hub
SESSION_IDLE_TIMEOUT=1800
```

```
DECSI_CUSTOMER_FALLBACK MOCK=True
DECSI_CBS_USE MOCK_LEDGER=True
CHAPA_FORCE MOCK=True
```

Generate SECRET_KEY (any machine with Python):

```
python3 -c "import secrets; print(secrets.token_urlsafe(50))"
```

Step 4 — Build and start

```
cd /opt/decsi_loan
docker compose up --build -d
docker compose ps
docker compose logs -f web
```

Wait until `db` is healthy and `web` is Up.

Step 5 — Migrate and create superuser

```
docker compose exec web python manage.py migrate
docker compose exec web python manage.py createsuperuser
docker compose exec web python manage.py check_license
```

Step 6 — Open the surfaces

Surface	URL
License status	<code>`http://<host>:8000/license/`</code>

Digital Apply	`http://<host>:8000/`
Staff hub	`http://<host>:8000/hub/login/`
Django Admin	`http://<host>:8000/admin/`
In-app Help	`http://<host>:8000/hub/help/`

Step 7 — Post-install configuration (functional go-live)

As superuser in the hub **Settings** (see Admin user manual):

1. Districts → Branches → geography
2. Staff users + roles + branch/district
3. Loan categories + document packs
4. Approval committees
5. Collateral / estimation mode
6. Digital Apply open/fee/security
7. MFA enrollment for privileged accounts
8. SMTP for password reset (if used)

Optional CSV imports:

```
docker compose exec web python manage.py import_zones <file>
docker compose exec web python manage.py import_branches <file>
docker compose exec web python manage.py import_loan_categories <file>
docker compose exec web python manage.py import_users <file>
```

5. HTTPS for field tablets (optional)

Camera / GPS often require HTTPS:

```
./scripts/gen_field_https_certs.sh <lan-ip>
docker compose -f docker-compose.yml -f docker-compose.https.yml
```

Use `https://<lan-ip>:8443` and update `SITE_URL /`
`CSRF_TRUSTED_ORIGINS` accordingly.

6. Smoke test checklist (IT acceptance)

#	Test	Expected
1	<code>`docker compose ps`</code>	<code>`web` Up, `db` healthy</code>
2	<code>`/license/`</code>	Valid · correct org · future expiry
3	<code>`/hub/login/`</code>	Login form (not license blocked page)
4	Superuser login	Hub home
5	<code>`/`</code>	Digital Apply landing
6	Create staff loan	Queue ID issued
7	Upload document	Stored under media
8	Digital Apply mock fee → submit	Appears in Online loan intake
9	Restart	<code>`docker compose restart`</code> — data persists

7. Backup and update

Backup

```
docker compose exec -T db pg_dump -U "$DB_USER" "$DB_NAME" > b
# Also back up Docker media volume and a secure offline copy c
```

Update

```
cd /opt/decsi_loan
# apply new release (git pull / unzip)
```

```
docker compose up -d --build
docker compose exec web python manage.py migrate
```

License keys survive updates if `.env` / `license.key` are preserved.

8. Troubleshooting

Symptom	Action
Page says “Product license inactive”	Check <code>`LICENSE_KEY`</code> , expiry, then <code>`check_license`</code>
<code>`DisallowedHost`</code>	Add host to <code>`ALLOWED_HOSTS`</code>
MFA codes fail	Fix server NTP / timezone
DB connection errors	Confirm <code>`DB_HOST=db`</code> , password, <code>`docker compose logs db`</code>
Need renewal	Email Seqela with org code + desired expiry

9. Support contacts

Role	Contact
Seqela product / license renewal	Your Seqela account manager (request renewal key with desired expiry)
DECSI IT operations	Local Dedebit technical team

Document control

Item	Value
Customer	Dedebit Credit and Savings Institution (DECSI)
Product	AI-powered Credit Intelligence / Loan Hub

Install	Docker Compose on-premises
License	`SEQLA1` Ed25519-signed key with expiry + grace
Companion	`docs/user_manual/06_installation_it.md`, user manuals PDF